

CALM VERITAS

The Vitamin B12 Guide

Deficiency, Absorption, and Dosing —
What the Research Actually Supports

Why B12 Matters

Vitamin B12 (cobalamin) is required for red blood cell formation, DNA synthesis, and maintenance of the myelin sheath that insulates nerve fibers. Unlike most water-soluble vitamins, B12 is stored in the liver in relatively large amounts — which is precisely why deficiency can take years to become symptomatic, and why it is often missed until nerve damage has already begun.

Who Is Most at Risk

— Adults over 50

Stomach acid production declines with age, and B12 absorption from food depends on adequate acid to release it from protein. Population studies estimate 10–30% of adults over 50 have reduced absorption capacity.

— Vegans and strict vegetarians

B12 is produced by bacteria and is reliably found only in animal products. Plant sources are unreliable or provide inactive analogues.

— People on metformin or long-term PPIs

Metformin is associated with reduced B12 absorption in an estimated 10–30% of long-term users, per multiple clinical studies. Proton pump inhibitors reduce the stomach acid needed for absorption.

— Those with pernicious anemia or gut conditions

An autoimmune condition that destroys intrinsic factor, the protein required to absorb B12 in the small intestine, unrelated to dietary intake.

Recognizing Deficiency

Early B12 deficiency symptoms are frequently mistaken for ordinary fatigue or aging, which delays diagnosis. A 2014 review in *American Family Physician* noted that neurological symptoms can precede blood-count abnormalities entirely, meaning a standard complete blood count can miss deficiency that is already affecting the nervous system.

Common Signs

- Persistent fatigue and weakness not explained by sleep or exertion
- Numbness or tingling in hands and feet (peripheral neuropathy)
- Difficulty concentrating or mild cognitive changes
- Glossitis (a smooth, sore tongue) and mouth ulcers
- Balance problems or unsteady gait in more advanced cases
- Mood changes, including depressive symptoms

Getting Tested

A serum B12 test is the standard first screen, but it has known limitations — levels can appear normal even with functional deficiency. Methylmalonic acid (MMA) testing is more sensitive: MMA rises when B12 is insufficient at the cellular level, even if serum B12 looks adequate. Ask a clinician about pairing both tests if deficiency is suspected despite a borderline-normal B12 result.

Forms & Dosing

Cyanocobalamin vs. Methylcobalamin

Cyanocobalamin is the most common and least expensive supplemental form; the body converts it to active forms after absorption. Methylcobalamin is already in an active, bioavailable form and is generally preferred for those with absorption or conversion concerns, though head-to-head evidence of clinical superiority is mixed. Both are effective for correcting dietary deficiency in people with normal absorption.

Recommended Daily Intake

The RDA for adults is 2.4 micrograms per day, rising to 2.6 mcg during pregnancy and 2.8 mcg while breastfeeding. For confirmed deficiency, clinical protocols typically use much higher doses — often 1,000 mcg daily or more — because absorption efficiency drops sharply as dose increases, particularly via the passive-diffusion pathway relied on at high oral doses.

Sublingual, Oral, or Injection?

For dietary insufficiency in otherwise healthy people, standard oral tablets are effective and inexpensive. For pernicious anemia or significant absorption impairment, high-dose oral (1,000–2,000 mcg/day) has been shown in multiple trials to be roughly as effective as intramuscular injection for most patients, though injections remain standard for severe or acute neurological deficiency. Sublingual forms have not been shown to meaningfully outperform standard high-dose oral tablets in controlled comparisons, despite common marketing claims.

Quick Reference

Summary Checklist

- 01 Know your risk factors: age 50+, plant-based diet, metformin, PPIs, gut surgery history
- 02 Request both serum B12 and MMA testing if deficiency is suspected
- 03 Choose methylcobalamin if absorption or conversion is a known concern
- 04 For confirmed deficiency, expect high-dose protocols (1,000+ mcg/day) rather than RDA-level doses
- 05 Recheck levels 3–4 months after starting supplementation to confirm correction
- 06 Don't rely on sublingual forms outperforming standard tablets — evidence doesn't support the claim

This guide is for educational purposes and does not replace individualized medical advice. Speak with a clinician before starting high-dose supplementation, particularly if you take other medications.